INTERNATIONAL GCSE MATHEMATICS A · MODULAR · UNIT 2 HIGHER

The Night-Before Revision Sheet

One page · one idea · one example

The 12 ideas in every Unit 2 paper (2025–2026). Read the key words, follow the example, then cover it and do it again.



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MARKS · ONE PAPER	QUESTIONS	IDEAS · 12 PAGES	EXAMPLE PER PAGE

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BOTTOM LINE Circle theorems need a reason for every angle. Vectors need a route written as letters before any **a** or **b**. Differentiation: set $\frac{dy}{dx} = 0$, then find y too.

1 Circle Theorems — With Reasons

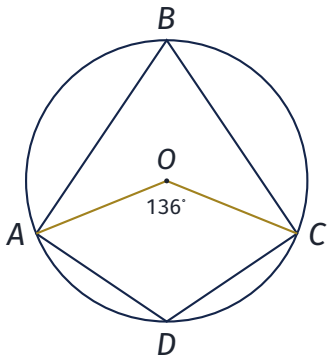
Find one angle at a time and write the theorem next to it.

WHEN YOU SEE → YOU DO

Angle at the centre and at the circumference	→ centre = 2× circumference
Four points on the circle (cyclic quadrilateral)	→ opposite angles add to 180°
Tangent and a chord	→ alternate segment: equals the angle in the other segment
Tangent and a radius	→ meet at 90°

EXAMPLE

A, *B*, *C* and *D* are points on a circle, centre *O*. Angle *AOC* = 136°.
Work out (a) angle *ABC* (b) angle *ADC*. Give a reason for each step.



WORKED SOLUTION

1 **(A) CENTRE IS DOUBLE THE CIRCUMFERENCE**
ABC = 136 ÷ 2 = 68° — *the angle at the centre is twice the angle at the circumference.*

2 **(B) CYCLIC QUADRILATERAL**
ADC = 180 − 68 = 112° — *opposite angles of a cyclic quadrilateral add up to* 180°.

ANSWER (a) 68° (b) 112°

TRAP Write the reason in words, not only the number — the mark needs it. Check which arc: *D* is on the same side as the 136° angle, so it is the obtuse one.

2 Vectors — a Route and a Proof

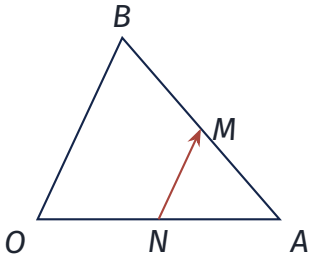
Write the route as letters first: $\vec{NM} = \vec{NO} + \vec{OM}$. Then swap in **a** and **b**.

WHEN YOU SEE → YOU DO

$\vec{AB}$ from $\vec{OA}$ and $\vec{OB}$	$\vec{AB} = \mathbf{b} - \mathbf{a}$
Midpoint M of AB	$\vec{OM} = \frac{1}{2}(\mathbf{a} + \mathbf{b})$
"Show that … is parallel to …"	one vector is a number × the other
"Show that P, Q, R lie on a straight line"	$\vec{PR} = k\vec{PQ}$ AND a common point

EXAMPLE

OAB is a triangle. $\vec{OA} = \mathbf{a}$ and $\vec{OB} = \mathbf{b}$. M is the midpoint of AB and N is the midpoint of OA.
Show that NM is parallel to OB.



WORKED SOLUTION

1. **ROUTE THROUGH O**
$\vec{NM} = \vec{NO} + \vec{OM}$
2. **EACH PIECE**
$\vec{NO} = -\frac{1}{2}\mathbf{a}$, $\vec{OM} = \frac{1}{2}(\mathbf{a} + \mathbf{b})$
3. **ADD**
$\vec{NM} = -\frac{1}{2}\mathbf{a} + \frac{1}{2}\mathbf{a} + \frac{1}{2}\mathbf{b} = \frac{1}{2}\mathbf{b}$

ANSWER $\vec{NM} = \frac{1}{2}\vec{OB}$, so NM is parallel to OB (and half its length).

TRAP Going against an arrow changes the sign. The conclusion sentence is a mark: say "a multiple of **b**, so parallel".

3 Functions — Composite and Inverse

$fg(x)$ means do g FIRST, then f. The inverse undoes the function.

WHEN YOU SEE → YOU DO

$fg(2)$	→ work out $g(2)$, then put the answer into f
$gf(x)$	→ replace every x in g with the whole of $f(x)$
$f^{-1}(x)$	→ $y = f(x)$, swap to make x the subject, rename
"which value must be excluded"	→ the value that makes a denominator 0

EXAMPLE

$f(x) = 3x - 2$ $\quad g(x) = \dfrac{1}{x+1}$

Find (a) $fg(2)$ (b) $gf(x)$ (c) $f^{-1}(x)$ (d) $g^{-1}(x)$

WORKED SOLUTION

1. **(A) g FIRST**
   $g(2) = \frac{1}{3}$, then $f\left(\frac{1}{3}\right) = 1 - 2 = -1$
2. **(B) PUT f INSIDE g**
   $$gf(x) = \frac{1}{(3x-2)+1} = \frac{1}{3x-1}$$
3. **(C) REARRANGE $y = 3x - 2$**
   $$x = \frac{y+2}{3}$$
4. **(D) REARRANGE $y = \dfrac{1}{x+1}$**
   $$x + 1 = \frac{1}{y} \Rightarrow x = \frac{1}{y} - 1$$

ANSWER (a) -1 (b) $\dfrac{1}{3x-1}$ (c) $f^{-1}(x) = \dfrac{x+2}{3}$ (d) $g^{-1}(x) = \dfrac{1-x}{x}$

TRAP $fg(x)$ is NOT $f(x) \times g(x)$. In the final answer of an inverse, write it in terms of x, not y.

4 Differentiation — Turning Points

Differentiate, set the gradient to zero, solve, then find y.

WHEN YOU SEE → YOU DO

$y = ax^n$	→ $\frac{dy}{dx} = nax^{n-1}$ (a number on its own goes)
"turning point", "maximum", "minimum"	→ $\frac{dy}{dx} = 0$
"gradient at $x = 2$"	→ substitute into $\frac{dy}{dx}$, not into y
"maximum or minimum?"	→ look at the shape, or the gradient just before and after

EXAMPLE

Find the coordinates of the turning points of $y = x^3 - 6x^2 + 9x + 1$ and say which is a maximum.

WORKED SOLUTION

1. **DIFFERENTIATE**
$\frac{dy}{dx} = 3x^2 - 12x + 9$
2. **SET EQUAL TO 0**
$3(x^2 - 4x + 3) = 0 \Rightarrow 3(x - 1)(x - 3) = 0 \Rightarrow x = 1$ or $x = 3$
3. **FIND y**
$x = 1$: $y = 1 - 6 + 9 + 1 = 5$ $x = 3$: $y = 27 - 54 + 27 + 1 = 1$
4. **WHICH IS WHICH**
A positive x^3 curve goes up, down, up: the first turning point is the maximum.

ANSWER maximum (1, 5), minimum (3, 1)

TRAP Give BOTH coordinates. The gradient is the $\frac{dy}{dx}$ value; the point is the (x, y) pair.

5 Simultaneous Equations — Line and Curve

Make y (or x) the subject of the LINE and substitute into the curve.

WHEN YOU SEE → YOU DO

One linear, one with x^2 or y^2	→	substitute the line into the curve
$x^2 + y^2 = r^2$	→	a circle, centre O, radius r
Quadratic in x appears	→	solve, then find each y from the LINE

EXAMPLE

Solve the simultaneous equations $y = 2x - 1$ $x^2 + y^2 = 13$

WORKED SOLUTION

1. **SUBSTITUTE**
   $x^2 + (2x - 1)^2 = 13$
2. **EXPAND AND COLLECT**
   $x^2 + 4x^2 - 4x + 1 = 13 \Rightarrow 5x^2 - 4x - 12 = 0$
3. **FACTORISE**
   $(5x + 6)(x - 2) = 0 \Rightarrow x = 2$ or $x = -\frac{6}{5}$
4. **y FROM THE LINE**
   $x = 2$: $y = 3$ $x = -\frac{6}{5}$: $y = -\frac{12}{5} - 1 = -\frac{17}{5}$

ANSWER $x = 2,\ y = 3$ or $x = -1.2,\ y = -3.4$

TRAP $(2x - 1)^2$ has a middle term: $4x^2 - 4x + 1$. Pair the answers: each x goes with its own y.

6 Cumulative Frequency — Median and IQR

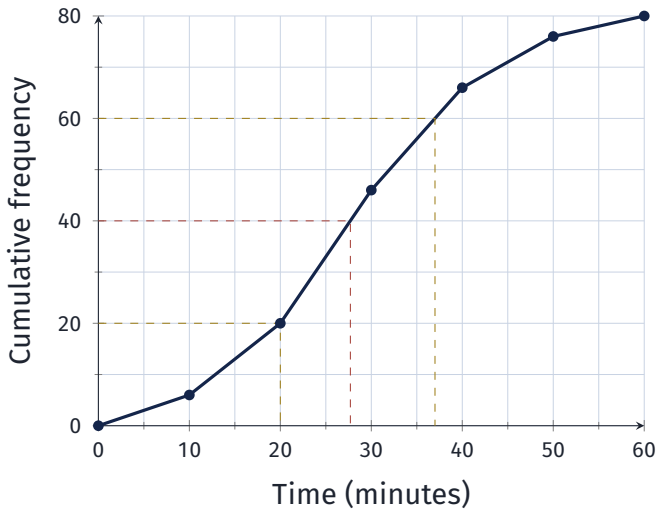
Read across from the cumulative frequency, then down to the value.

WHEN YOU SEE → YOU DO

Median (n people)	→	read at $\frac{n}{2}$
Lower / upper quartile	→	read at $\frac{n}{4}$ and $\frac{3n}{4}$; IQR = UQ − LQ
"how many more than 45"	→	read up from 45, then total − that reading

EXAMPLE

The graph shows the times, in minutes, that 80 people waited. Find estimates for the median and the interquartile range.



WORKED SOLUTION

1. **MEDIAN AT 40**
   read across from 40: about 28 minutes
2. **QUARTILES AT 20 AND 60**
   LQ = 20, UQ ≈ 37
3. **IQR**
   37 − 20 = 17

ANSWER median ≈ 28 min, IQR ≈ 17 min

TRAP Plot cumulative frequency at the END of each class. Read the quartiles at 20 and 60 — not at 25% of the time axis.

7 Percentages — Reverse and Compound

Turn every percentage change into a multiplier.

WHEN YOU SEE → YOU DO

Increase by 3%	⟶	×1.03
Decrease (depreciate) by 15%	⟶	×0.85
"original price", "before the sale"	⟶	DIVIDE by the multiplier
"each year for 4 years"	⟶	multiplier to the power 4

EXAMPLE

(a) In a sale, prices are reduced by 15%. A jacket costs \$68 in the sale. Work out its price before the sale.
(b) \$5000 is invested at 3% compound interest per year. Work out its value after 4 years.

WORKED SOLUTION

1. **(A) SALE PRICE** = 0.85× **ORIGINAL**
   original = 68 ÷ 0.85
2. **(B) FOUR YEARS AT** ×1.03
   $5000 \times 1.03^4 = 5627.544\ldots$

ANSWER (a) \$80 (b) \$5627.54

TRAP Never add 15% of the sale price back on: 68 + 10.20 = 78.20 is wrong. Compound interest is not 4 × 3% = 12%.

8 Similar Shapes — Area and Volume

Find the LENGTH scale factor first; then square it for area, cube it for volume.

WHEN YOU SEE → YOU DO

Length scale factor k	⟶	area $\times k^2$, volume $\times k^3$
Areas given	⟶	$k = \sqrt{\text{area ratio}}$
Volumes given	⟶	$k = \sqrt[3]{\text{volume ratio}}$

EXAMPLE

Two solid cones are mathematically similar. Their surface areas are 45 cm^2 and 180 cm^2. The volume of the smaller cone is 30 cm^3. Work out the volume of the larger cone.

WORKED SOLUTION

1. **AREA SCALE FACTOR**
   $180 \div 45 = 4$
2. **LENGTH SCALE FACTOR**
   $\sqrt{4} = 2$
3. **VOLUME SCALE FACTOR**
   $2^3 = 8$

ANSWER $30 \times 8 = 240\ cm^3$

TRAP Do not multiply the volume by 4 (the area factor). Always go back to the length factor first.

9 Direct and Inverse Proportion

Write the equation with k, find k, then use it.

WHEN YOU SEE → YOU DO

y is directly proportional to x^2	$\longrightarrow\ y = kx^2$
y is inversely proportional to x^2	$\longrightarrow\ y = \dfrac{k}{x^2}$
"proportional to the square root"	$\longrightarrow\ \sqrt{x}$ in the formula

EXAMPLE

y is inversely proportional to x^2. When $x = 2$, $y = 12$.
(a) Find a formula for y in terms of x. (b) Find y when $x = 4$. (c) Find the positive value of x when $y = 0.75$

WORKED SOLUTION

1. **FIND k**
$12 = \dfrac{k}{2^2} \Rightarrow k = 48$

2. **(B)**
$y = \dfrac{48}{16} = 3$

3. **(C)**
$0.75 = \dfrac{48}{x^2} \Rightarrow x^2 = 64 \Rightarrow x = 8$

ANSWER (a) $y = \dfrac{48}{x^2}$ (b) 3 (c) 8

TRAP The formula must include the value of k — "$y = k/x^2$" alone is not the answer.

10 Cones, Spheres and Pyramids

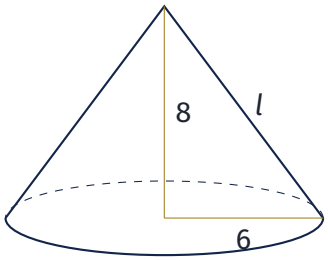
Use the formulae sheet — but find the slant height or radius first.

WHEN YOU SEE → YOU DO

Cone: volume	→ $\frac{1}{3}\pi r^2 h$
Cone: curved surface	→ $\pi r l$, with $l = \sqrt{r^2 + h^2}$
Sphere / hemisphere	→ $\frac{4}{3}\pi r^3$, $4\pi r^2$; a hemisphere is half — plus its flat circle
"in terms of π"	→ leave π in the answer

EXAMPLE

A solid cone has base radius 6 cm and height 8 cm. Work out (a) its volume and (b) its total surface area. Give your answers in terms of π.



WORKED SOLUTION

1. **VOLUME**
   $\frac{1}{3}\pi(6^2)(8) = 96\pi$
2. **SLANT HEIGHT**
   $l = \sqrt{6^2 + 8^2} = 10$
3. **SURFACE = CURVED + BASE**
   $\pi(6)(10) + \pi(6^2) = 60\pi + 36\pi$

ANSWER (a) 96π cm^3 (b) 96π cm^2

TRAP The curved surface uses the slant height l, not the height h. "Total" surface area includes the base.

11 Inequalities — Quadratic and Regions

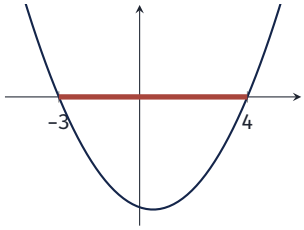
For a quadratic inequality: solve the equation, sketch, then read the part you want.

WHEN YOU SEE → YOU DO

$x^2 - x - 12 < 0$	⟶	between the roots
$x^2 - x - 12 > 0$	⟶	outside the roots: two parts
Region on a graph	⟶	one inequality for each line; test a point inside
Dashed line / solid line	⟶	< or > / ≤ or ≥

EXAMPLE

Solve $x^2 - x - 12 < 0$



WORKED SOLUTION

1. **SOLVE THE EQUATION**
   $(x - 4)(x + 3) = 0 \Rightarrow x = 4$ or $x = -3$
2. **SKETCH**
   a U-shape: below the axis BETWEEN the roots

ANSWER $-3 < x < 4$

TRAP Do not write $x < 4$ and $x < -3$. For "> 0" the answer is $x < -3$ or $x > 4$.

12 Arithmetic Sequences and Series

Same difference each time: nth term = $dn + (a - d)$, sum = $\frac{n}{2}[2a + (n - 1)d]$.

WHEN YOU SEE → YOU DO

nth term of 7, 11, 15, …	⟶ $4n + 3$
"Is 150 a term?"	⟶ solve $4n + 3 = 150$: whole number or not?
"Sum of the first 30 terms"	⟶ $S_n = \frac{n}{2}[2a + (n - 1)d]$ (on the formulae sheet)
Two terms given	⟶ two equations in a and d

EXAMPLE

The sequence 7, 11, 15, 19, … is arithmetic.
(a) Find the nth term. (b) Is 150 a term? Give a reason. (c) Work out the sum of the first 30 terms.

WORKED SOLUTION

1. **(A)**
   $d = 4$, so $4n + 3$
2. **(B)**
   $4n + 3 = 150 \Rightarrow n = 36.75$ — not a whole number
3. **(C)**
   $S_{30} = \frac{30}{2}[2(7) + 29(4)] = 15 \times 130$

ANSWER (a) $4n + 3$ (b) No (c) 1950

TRAP a is the first term (7), not the difference. In S_n it is $(n - 1)d$, not nd.